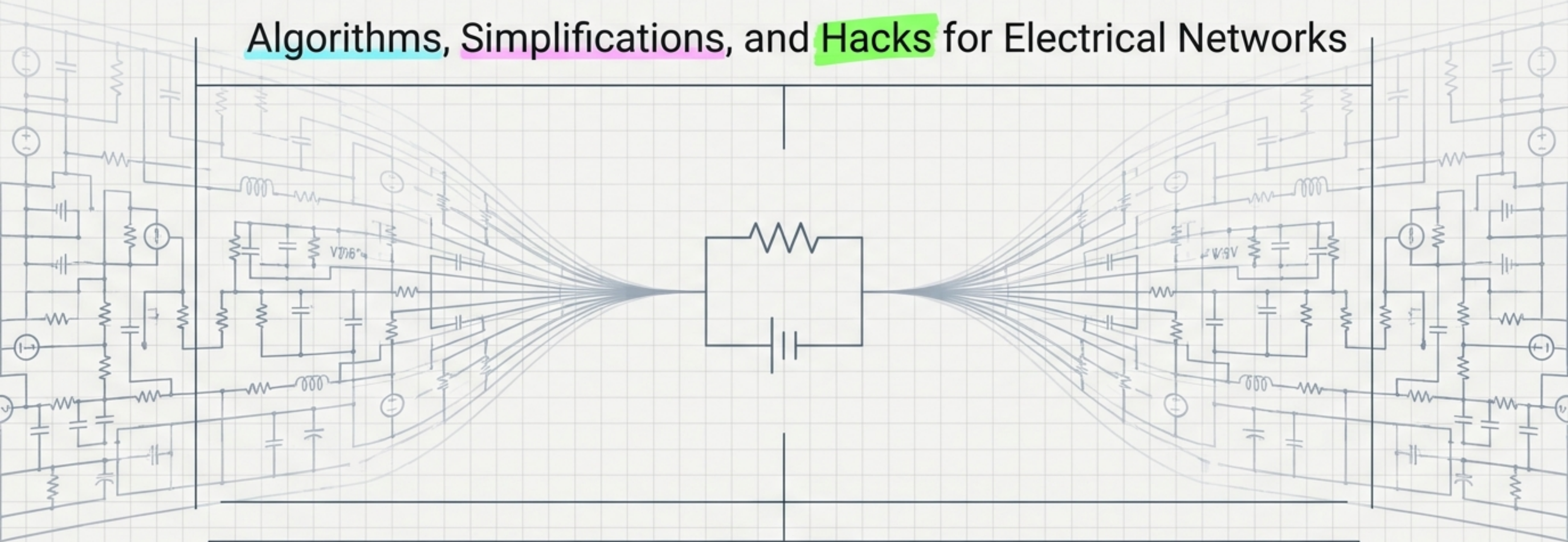


The Circuit Solver's Toolkit

Algorithms, Simplifications, and Hacks for Electrical Networks



EC1101 Master Reference Guide

The Physical Intuition: $V = I \times R$



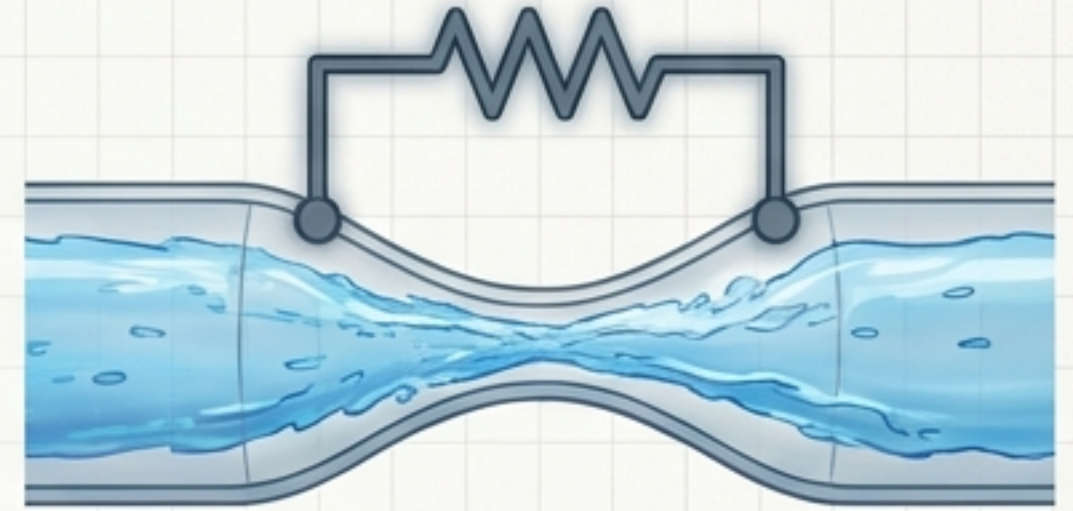
Voltage (V)

The pressure pushing the water along the pipe.



Current (I)

How much water actually flows past you per second.



Resistance (R)

How much the pipe fights back and narrows the flow.

Squeeze the pipe and the flow drops. Push harder and the flow rises. That is Ohm's Law in one sentence.

The Mathematical Engine: Cramer's Rule

Step 1: Cast to Matrix. Structure simultaneous equations into $\mathbf{AX} = \mathbf{B}$.

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + a_{13}x_3 &= b_1 \\ a_{21}x_1 + a_{22}x_2 + a_{23}x_3 &= b_2 \\ a_{31}x_1 + a_{32}x_2 + a_{33}x_3 &= b_3 \end{aligned} \rightarrow \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

$\mathbf{A} \qquad \mathbf{X} \qquad \mathbf{B}$

Step 2: The Denominator (Δ). Find the determinant of matrix $\mathbf{A}$.

$$\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

$$\begin{vmatrix} a_{11} & a_{12} & a_{13} & a_{11} & a_{12} \\ a_{21} & a_{22} & a_{23} & a_{21} & a_{22} \\ a_{31} & a_{32} & a_{33} & a_{31} & a_{32} \end{vmatrix}$$

$$\Delta = (a_{11}a_{22}a_{33} + a_{12}a_{23}a_{31} + a_{13}a_{21}a_{32}) - (a_{31}a_{22}a_{13} + a_{32}a_{23}a_{11} + a_{33}a_{21}a_{12})$$

Step 3: The Column Swap. Replace the k -th column of $\mathbf{A}$ with column $\mathbf{B}$ to find Δ_k .

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow \Delta_2 = \begin{vmatrix} a_{11} & b_1 & a_{13} \\ a_{21} & b_2 & a_{23} \\ a_{31} & b_3 & a_{33} \end{vmatrix}$$

$\mathbf{B}$

Step 4: Solve. $x_k = \Delta_k / \Delta$.

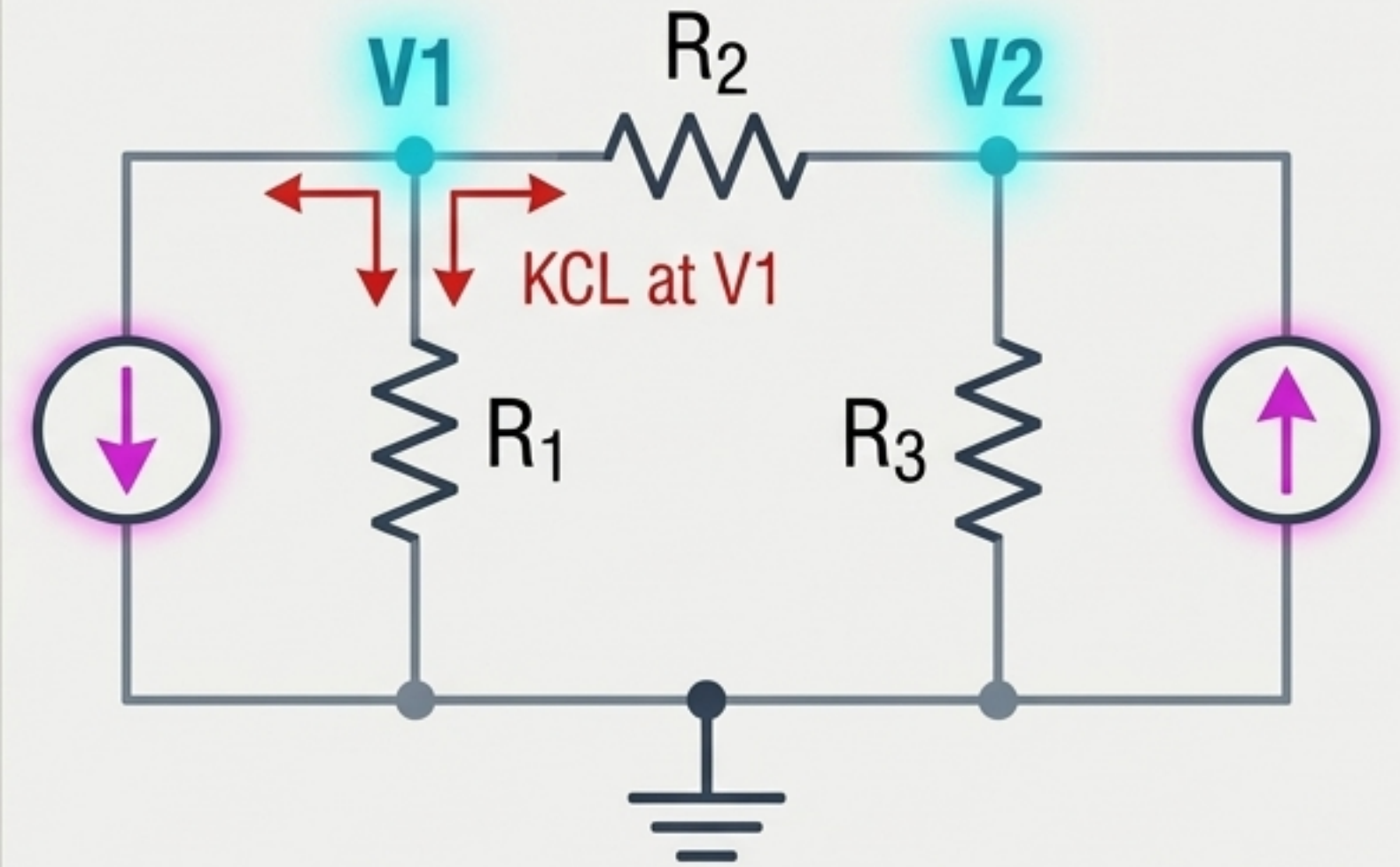
$$x_k = \frac{\Delta_k}{\Delta}$$
$$x_1 = \frac{\Delta_1}{\Delta}, x_2 = \frac{\Delta_2}{\Delta}, x_3 = \frac{\Delta_3}{\Delta}$$

Mathematical Dead End: If $\Delta = 0$, the equations are linearly dependent. There is no unique solution.

Method 1: Nodal Analysis

Applying KCL to find unknown node voltages.

1. **Ground it.** Select a reference node (Ground = 0V). Assign voltages (V_1 , V_2 , ... V_{n-1}) to the rest.
2. **Sum the currents.** Apply KCL at each non-reference node. (Current In = Current Out).
3. **Convert to Volts.** Use Ohm's Law to express branch currents as $(V_{\text{start}} - V_{\text{end}}) / R$.

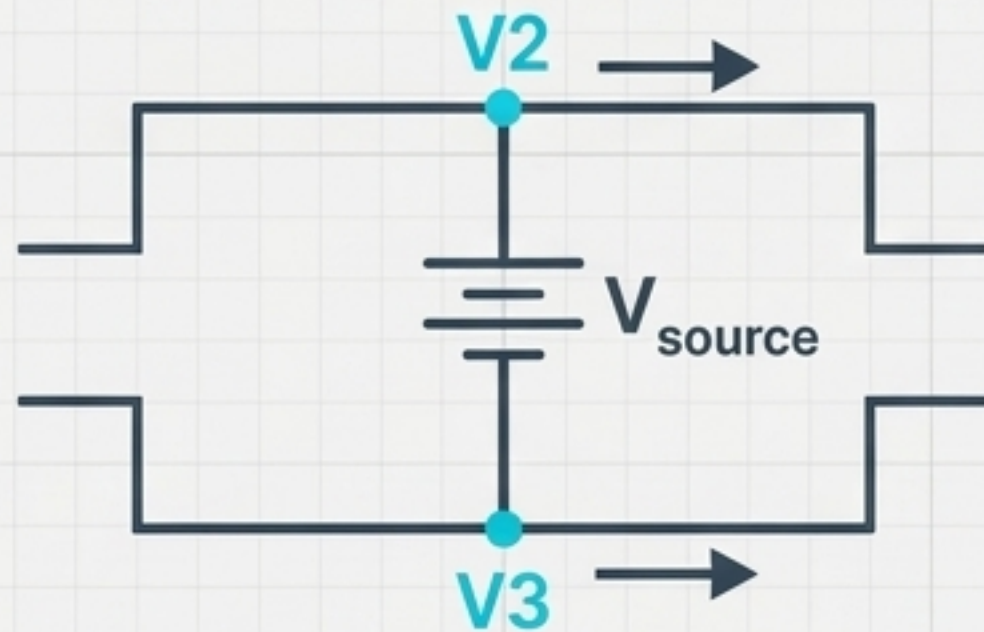


Mistake to Avoid: Forgetting to subtract the destination node voltage when expressing current (e.g., writing V_1/R_2 instead of $(V_1 - V_2)/R_2$).

Method 1b: The Supernode Hack

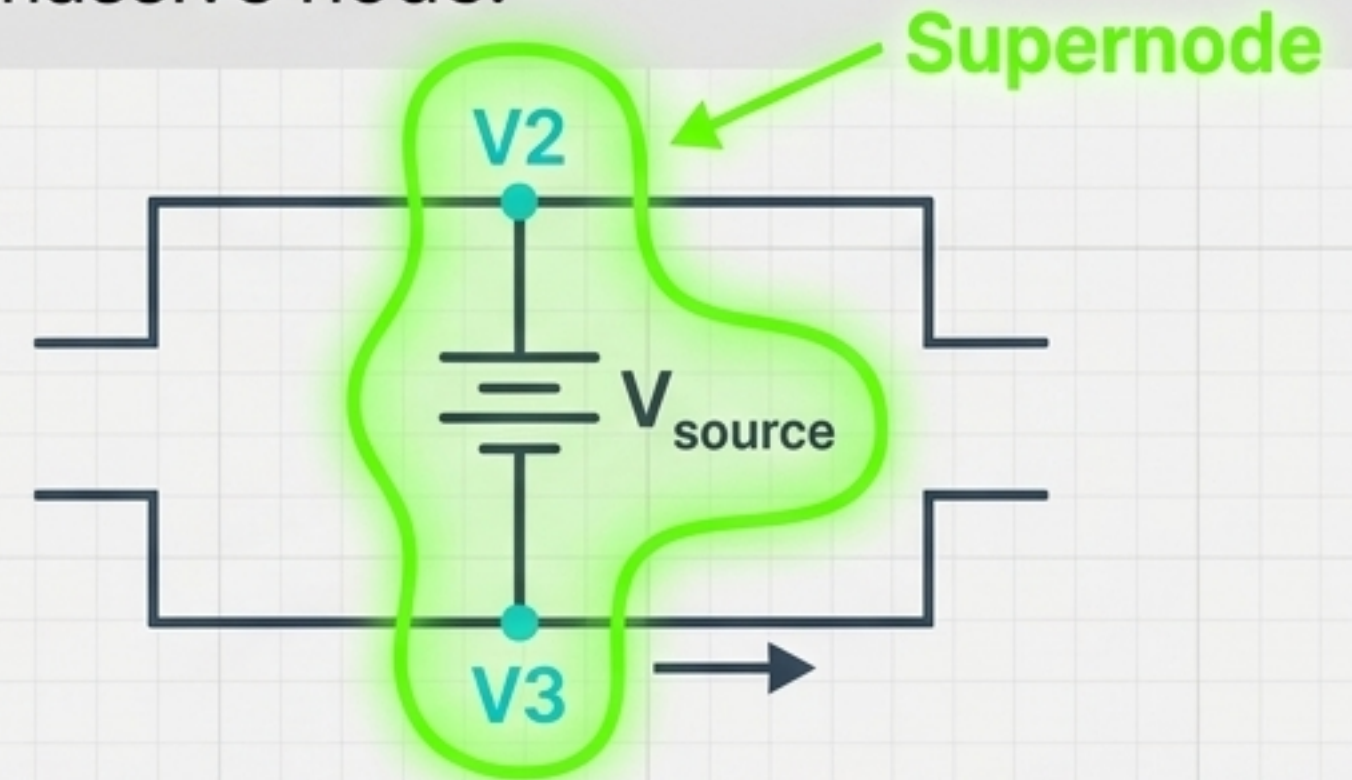
The Obstacle:

A voltage source sits between two non-reference nodes. You don't know the current through it, so KCL breaks.



The Hack:

Draw a boundary around the source and both nodes. Treat the entire boundary as one massive node.



Rules:

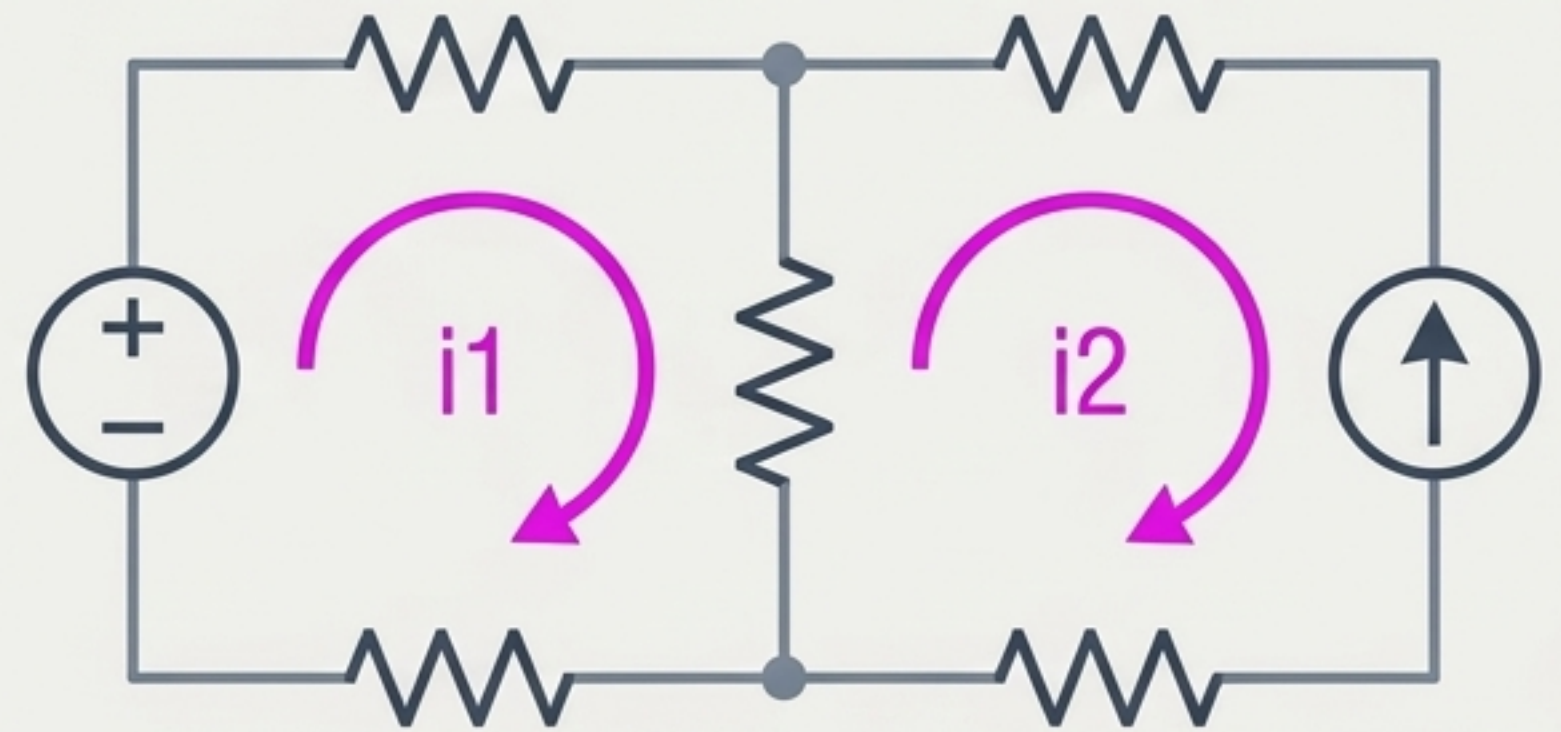
Rule 1: Apply KCL to the new Supernode (Total Current In = Total Current Out).

Rule 2: Create a constraint equation for the inside: $V_{\text{positive}} - V_{\text{negative}} = V_{\text{source}}$.

Method 2: Mesh Analysis

Applying KVL to find closed-loop mesh currents.

1. **Assign Loops.** Identify independent meshes (window panes). Assign a clockwise current (i_1 , i_2) to each.
2. **Walk the loop.** Apply KVL around each mesh. The sum of voltage drops must equal zero.
3. **Handle Shared Borders.** For a resistor shared by two meshes, the current is the difference between the two loops (e.g.,

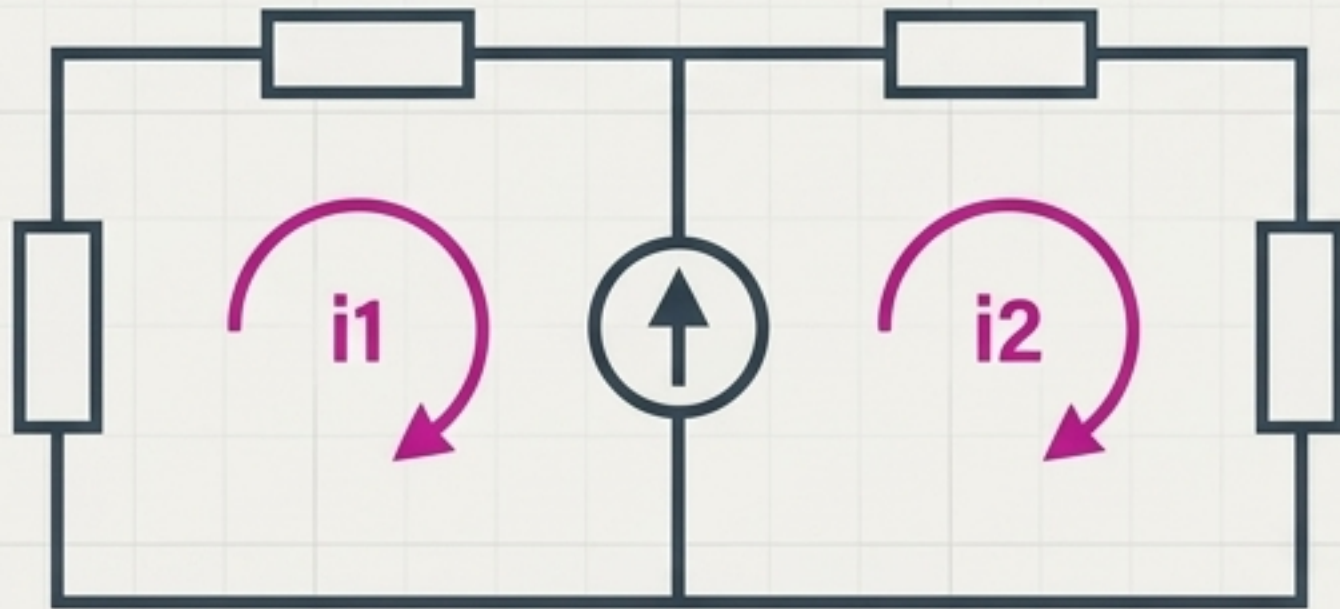


Mistake to Avoid: Getting signs wrong on shared resistors. Always prioritize the current of the mesh you are currently analyzing (e.g., $(i_1 - i_2)$ for Mesh 1, $(i_2 - i_1)$ for Mesh 2).

Method 2b: The Supermesh Hack

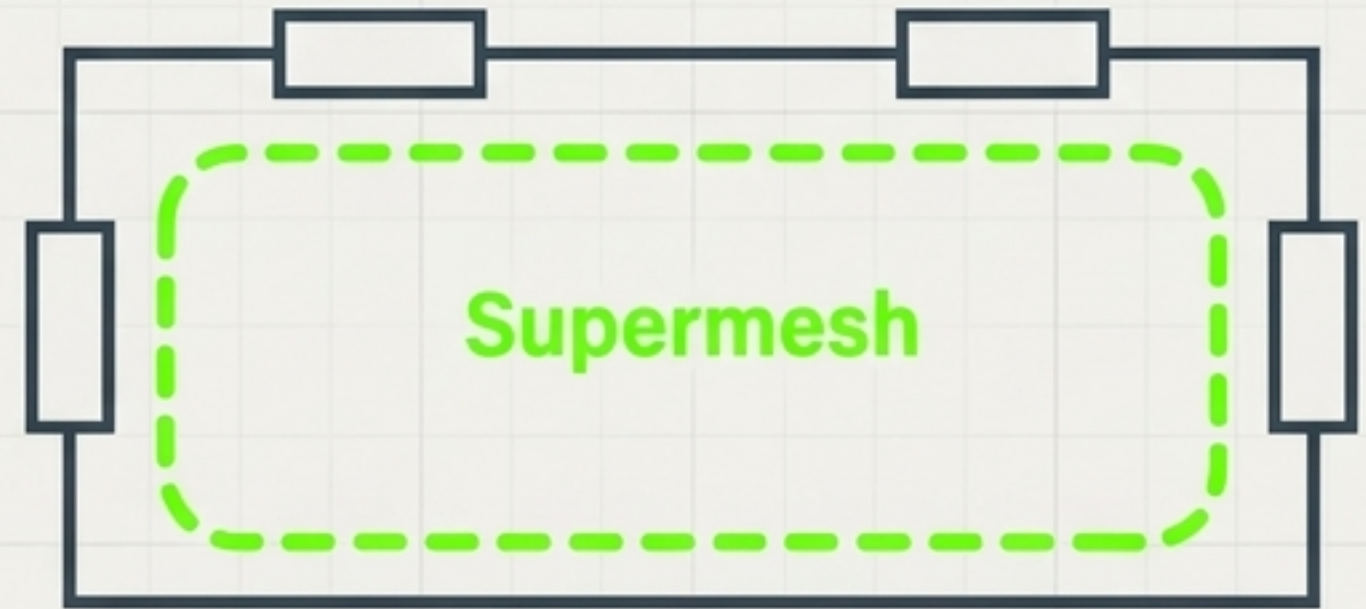
The Obstacle:

A current source sits on the border between two meshes. You don't know the voltage across it, so KVL breaks.



The Hack:

Temporarily remove the current source and the shared branch. Walk KVL around the larger combined outer loop.



Rules:

Rule 1: Apply KVL to the massive outer Supermesh.

Rule 2: Create a constraint equation for the shared border: $i_{\text{mesh}} - i_{\text{adjacent}} = I_{\text{source}}$.

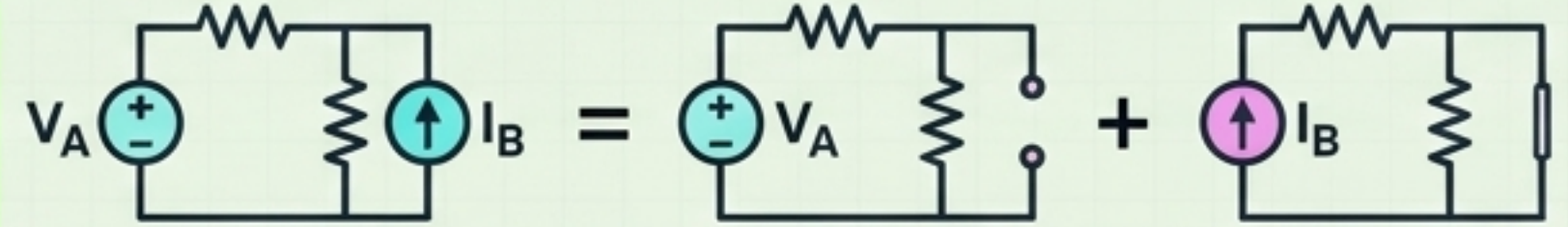
Synthesis: Choosing the Right Engine

	Nodal Analysis	Mesh Analysis
Law Used (The Physics)	KCL (Currents at a point)	KVL (Voltages around a loop)
Target (The Variable)	Node Voltages (V_n)	Mesh Currents (i_n)
Villain (The Obstacle)	Floating Voltage Source	Shared Current Source
Solution (The Hack)	Supernode	Supermesh
Topology (When to use)	Best for circuits with many parallel branches and current sources.	Best for planar circuits with many series components and voltage sources.

Simplification 1: Superposition

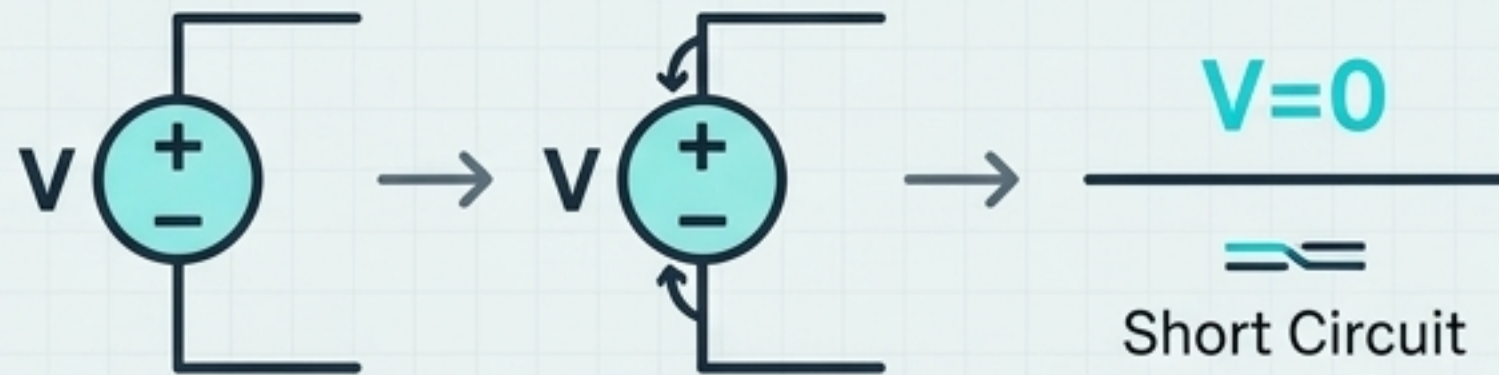
Divide, conquer, and algebraically sum.

The Concept: In a linear circuit, the total response is the sum of responses caused by each independent source acting alone.

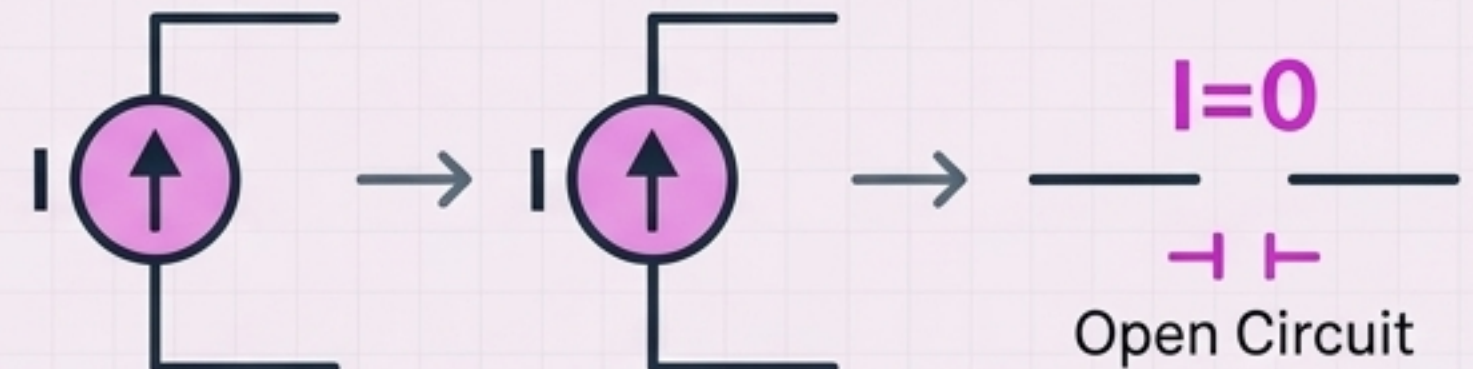


The Kill Switches: Turning sources OFF to isolate variables.

Voltage Sources → Short Circuit
($V = 0$ means a piece of wire).



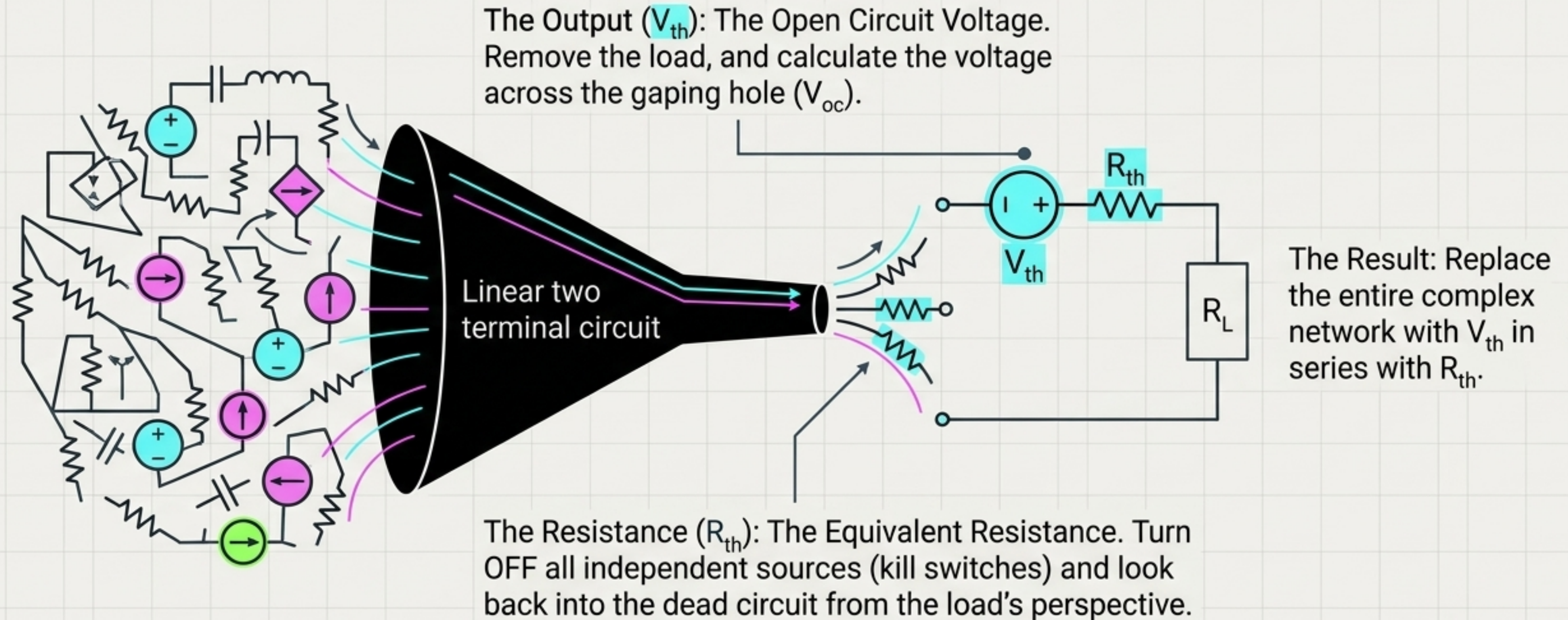
Current Sources → Open Circuit
($I = 0$ means infinite resistance).



Mistake to Avoid: Power ($P = I^2R$) is NOT linear. You cannot superimpose power calculations directly; you must find total current/voltage first.

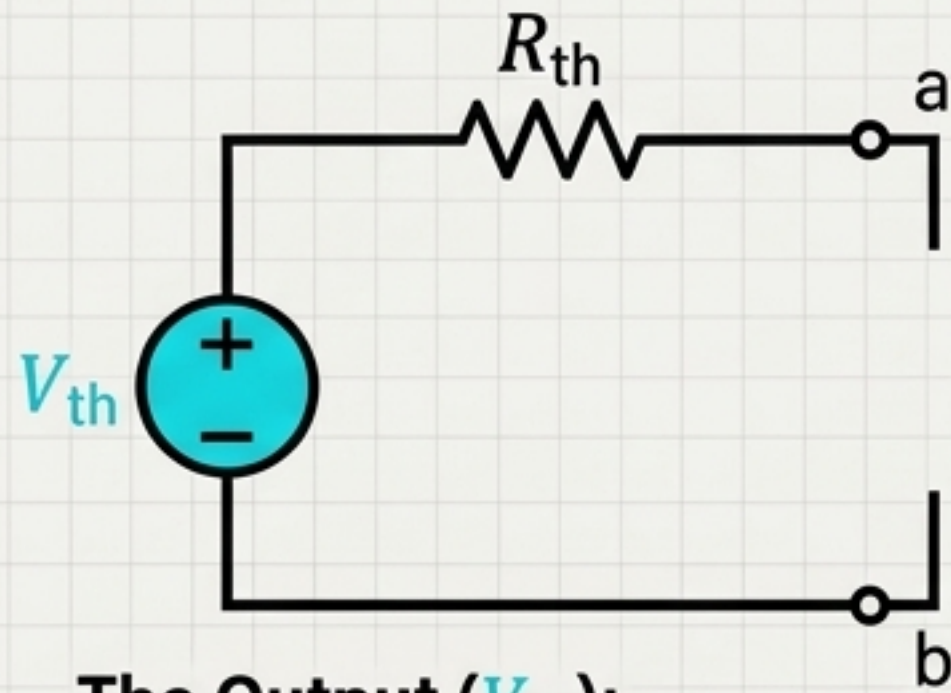
Simplification 2: Thevenin's Theorem

Collapsing the entire universe into one battery and one resistor.



Simplification 3: Norton's Theorem

The current-source mirror of Thevenin.

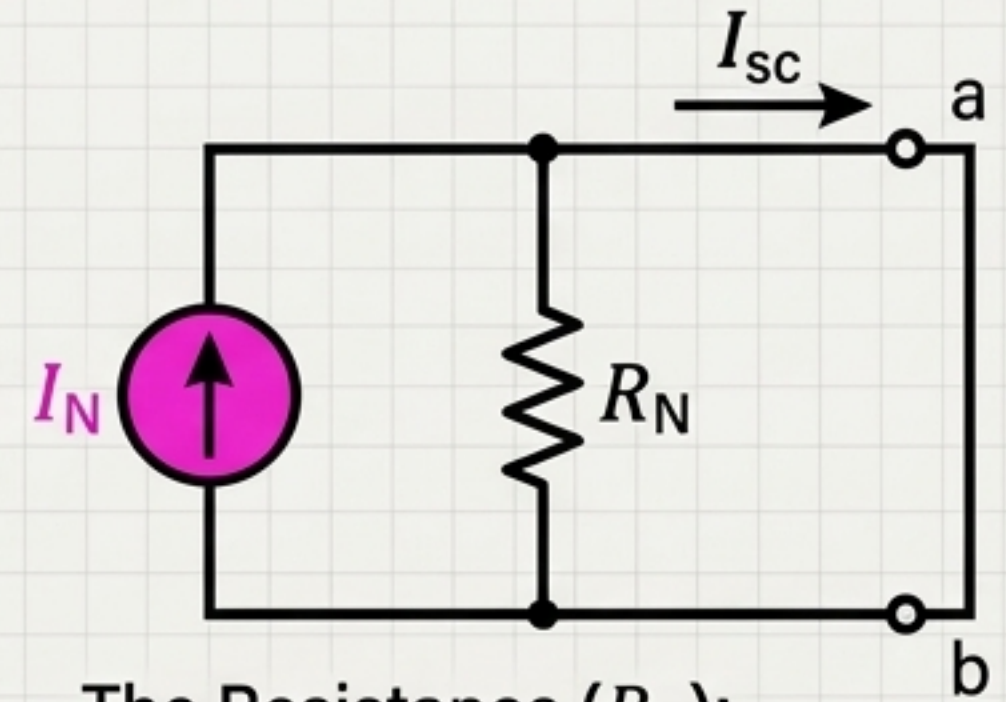


The Output (V_{th}):
The Open Circuit Voltage.



Source Transformation:

Source Transformation:
You can freely swap between
Thevenin and Norton models using
Ohm's Law: $V_{th} = I_N \times R_{th}$.



The Resistance (R_N):
Exactly the same as R_{th} .

The Output (I_N): The Short Circuit Current.
Replace the load with a solid wire,
and calculate the current flowing
through that specific wire (I_{sc}).

Mistake to Avoid: In Norton, the resistor R_N goes in **PARALLEL** with the current source. Do not put it in series.

The Red Flag: Dependent Sources

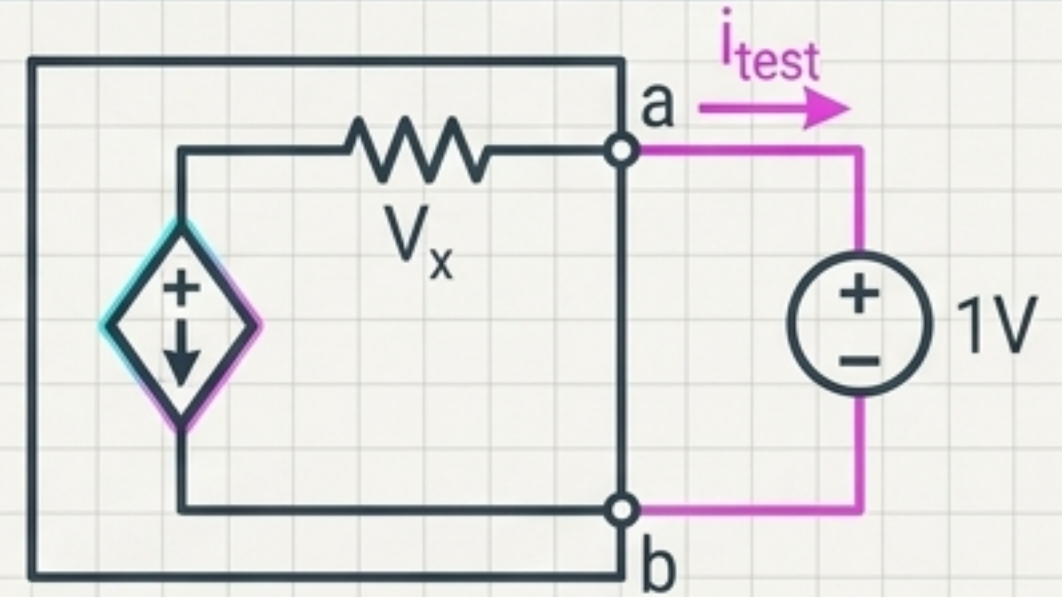
They are active observers. You **CANNOT** turn them off.



**Kill
Independent
Sources**



**Leave
Dependent
Sources
ACTIVE**



How to find R_{th} with Dependent Sources:

1. Remove the load.
2. Turn off all independent sources (circles).
3. Apply a test source (1V or 1A) at the open terminals.
4. Calculate resulting current (i_{test}) or voltage (V_{test}).
5. $R_{th} = V_{test} / i_{test}$.



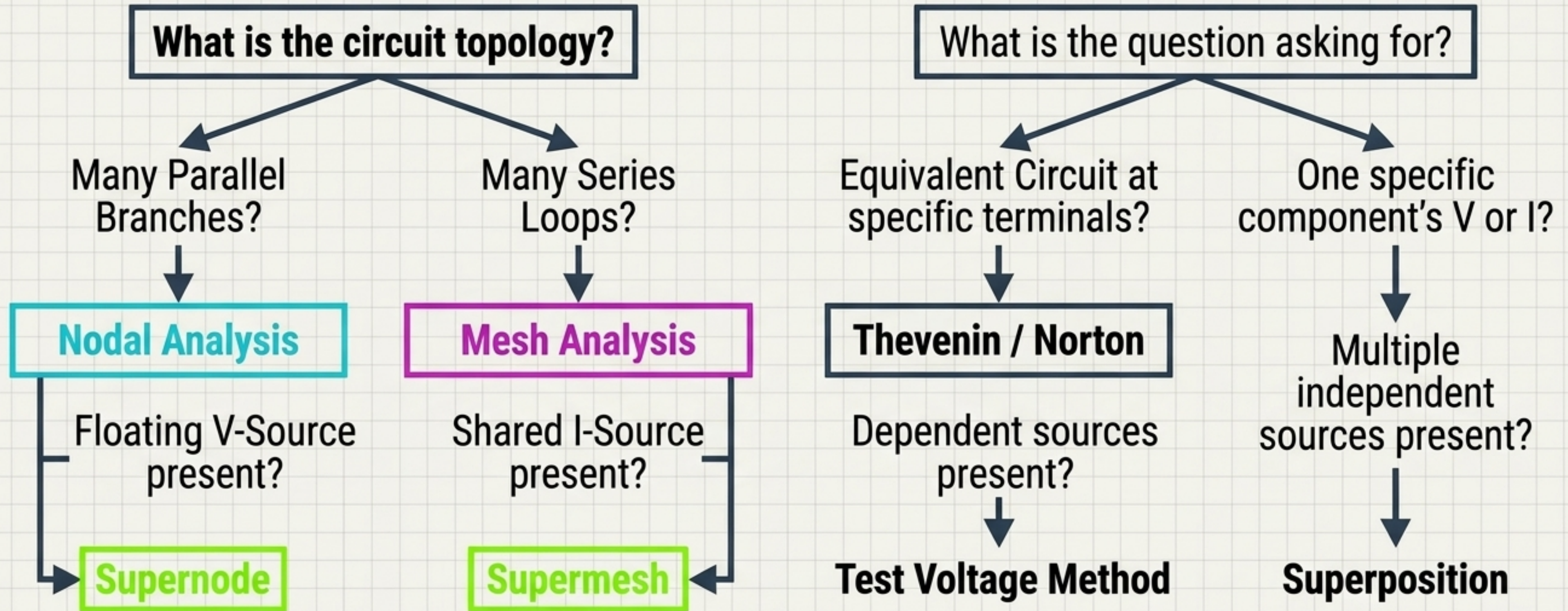
Wait, is R_{th} negative?

A negative R_{th} is **not a math error**. It means the dependent source is acting as an **active device supplying power** to the circuit!

Synthesis: The Network Theorems

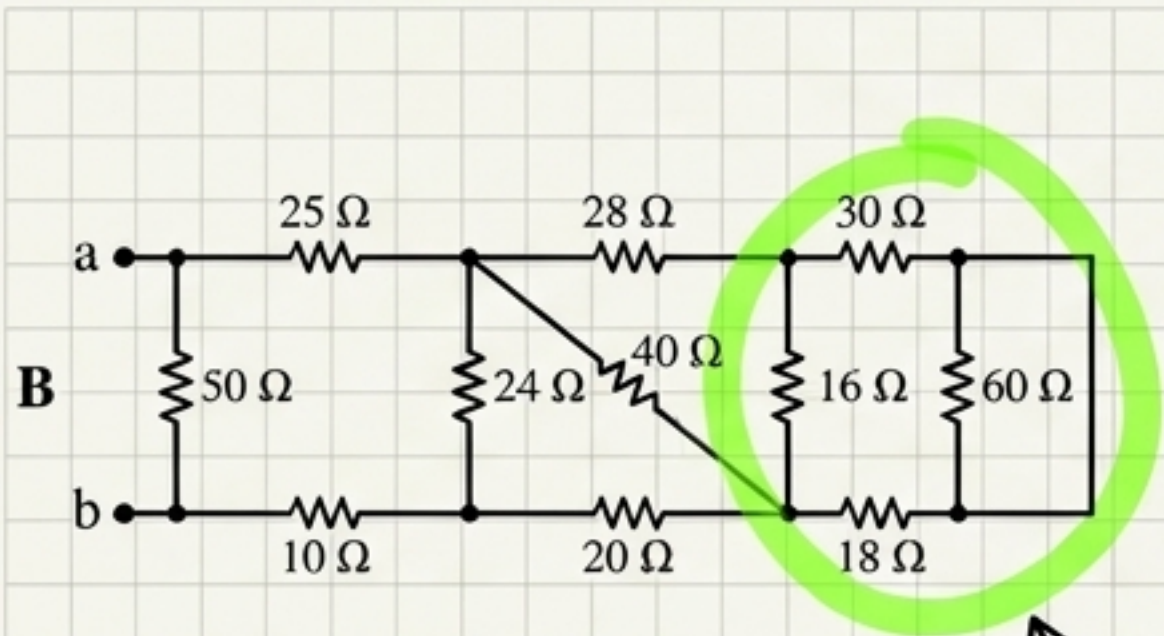
	Thevenin	Norton
Model Structure	Voltage Source in Series	Current Source in Parallel
Target Calculation	Open Circuit Voltage (V_{oc})	Short Circuit Current (I_{sc})
Shared Metric	R_{th} and R_N are mathematically identical. (Turn sources off, look into terminals).	
The Link	$V_{th} = I_N \times R_{th}$	$I_N = V_{th} / R_N$

The Master Decision Tree

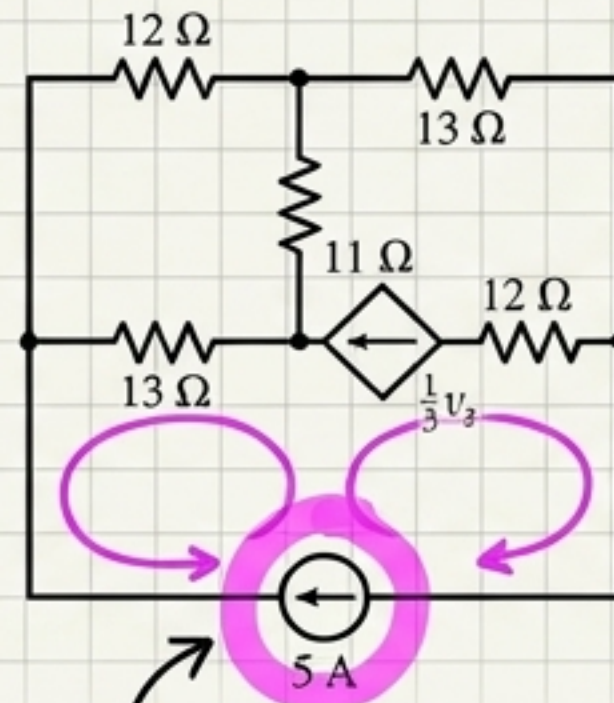


COMMON PITFALL: Don't mix Nodal and Mesh equations;
pick ONE and stick to it!

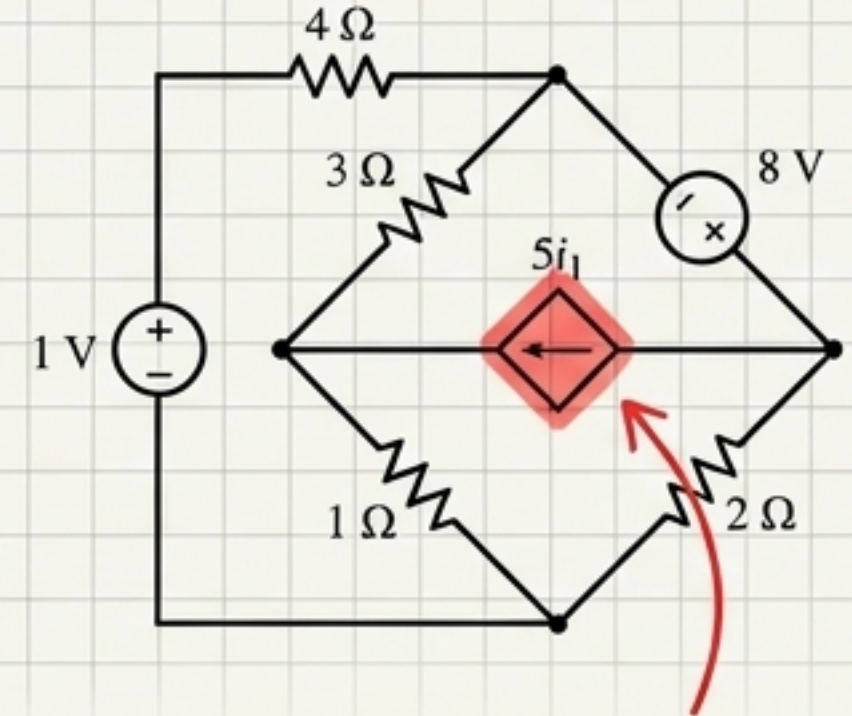
Applying the Toolkit



Pure Ohmic reduction.
Work backwards from the
furthest point.



Classic Supermesh trigger.



Red flag! R_{th} requires a
 1 V test source. Do NOT
turn this off.

The circuit doesn't care which method you use, but your time limit does.
Read the blueprint before you build.